How would an economist solve climate change?

ECON-4210

Economics of Climate Change

Class 15

Prof. Casey Wichman

Georgia Institute of Technology

Today's class

1. Questions?
2. Attendance
3. Housekeeping
    1. Unit 3 overview
    2. Ghost Emissions round 6
4. **How would an economist solve climate change?**
    1. Planet Money (2020), "Sell Me Your Climate Bombs." Podcast.
5. For next class…

Unit 3: Climate policy

- How would an economist solve climate change? (classes 15–16)
- Climate policy in practice: carbon pricing and its politics (class 17)
- In-class policy game (class 18)
- Who pays for a carbon tax? (class 19)
- Does behavior undermine climate policy? Rebound effects (class 20)
- How do we cut emissions cheaply? Innovation and learning curves (class 21)

- ***Problem set 3 due:*** Friday, 11/13, 11:59 PM.
- ***Exam 3:*** Monday, 11/16 (classes 15–21).

Ghost Emissions, round 6: Pick a tool

- Each borough picks one policy tool this week.
- **Businesses/Mayors** submit by Sunday, 11:59 PM.

- **Tax:** businesses pay τ for every ton of ghost emissions released.
  - The Pigouvian tax from today.
- **Cap:** a fixed number of permits, handed out by each business's baseline share of emissions and traded within the borough.
  - Cap-and-trade with free permits.
- **Minimum-containment rule:** every business contains the same share of its emissions.
  - A performance standard.

- **Question to consider:** With known costs, which two tools produce the same containment? Under each tool, who benefits the most?

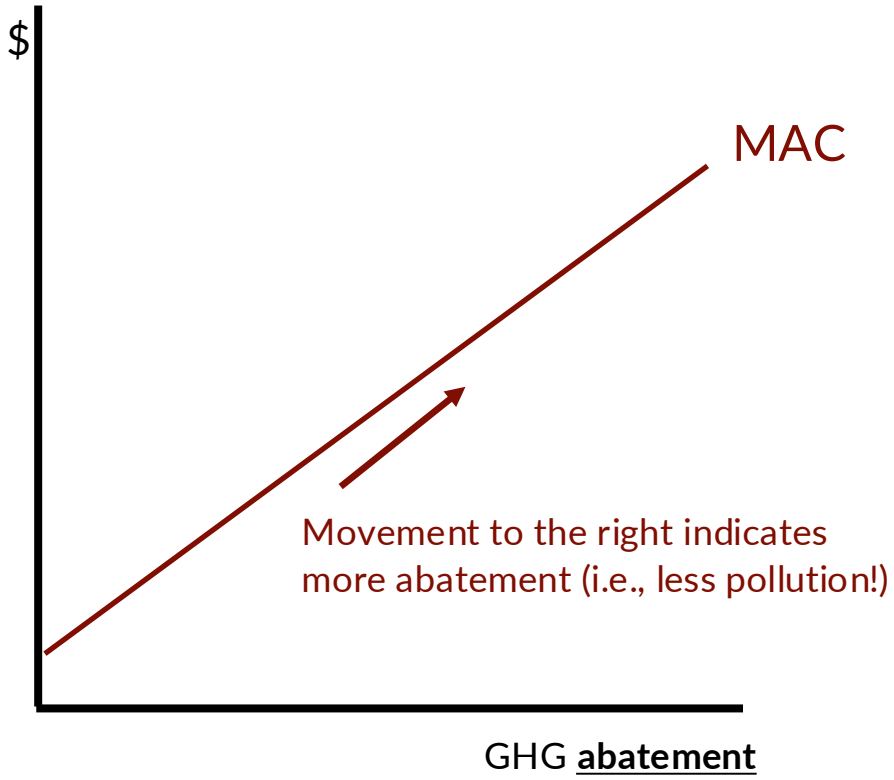
Mechanisms to control GHG emissions

- Environmental policy has largely relied on direct regulation or **"command and control (CAC)"** policies
  - **Performance standard:** government specifies the amount of pollution allowed from each firm
  - **Technology standard:** firms are required to use particular method or technology to reduce emissions

- **Market-based approaches:**
  - **Pigouvian (or corrective) taxes: government sets tax equal to marginal damages.**
  - **Cap-and-trade: government caps total emissions and allows firms to trade the "right" to emit GHGs.**

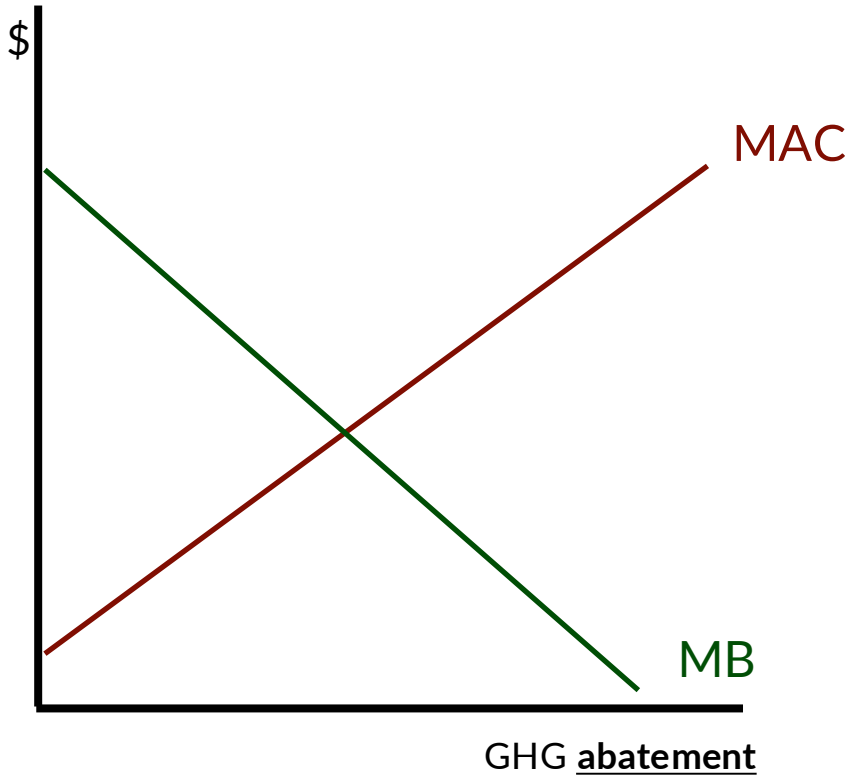


Useful to think in terms of GHG *abatement*



- Abatement costs are the costs of *abating* or reducing pollution.
- **Marginal abatement costs (MACs)** are the costs of reducing each additional emission of a pollutant.
  - Reflects minimum cost of achieving each unit of abatement (i.e., do the cheapest thing first!)
- Ways firms can reduce pollution (all else equal):
  - Produce less output
  - Switch technology (e.g., use "scrubbers")
  - Use cleaner inputs (e.g., NG rather than coal)
  - →movement along the MAC curve

Useful to think in terms of GHG *abatement*



- **Marginal benefits (MB)** are the marginal benefits of reducing each additional emission of a pollutant.
  - This is the same idea we've been working with so far!
  - *This MB curve doesn't exist in the real world!*

- Think of a global damage function.
  - Damage $= 0.283T + 0.146T^2$
  - $\Longrightarrow$ MD $= 0.283 + 0.292T$
  - But more GHG abatement reduces T, so slope of **MB** would be negative. (MB = –MD).

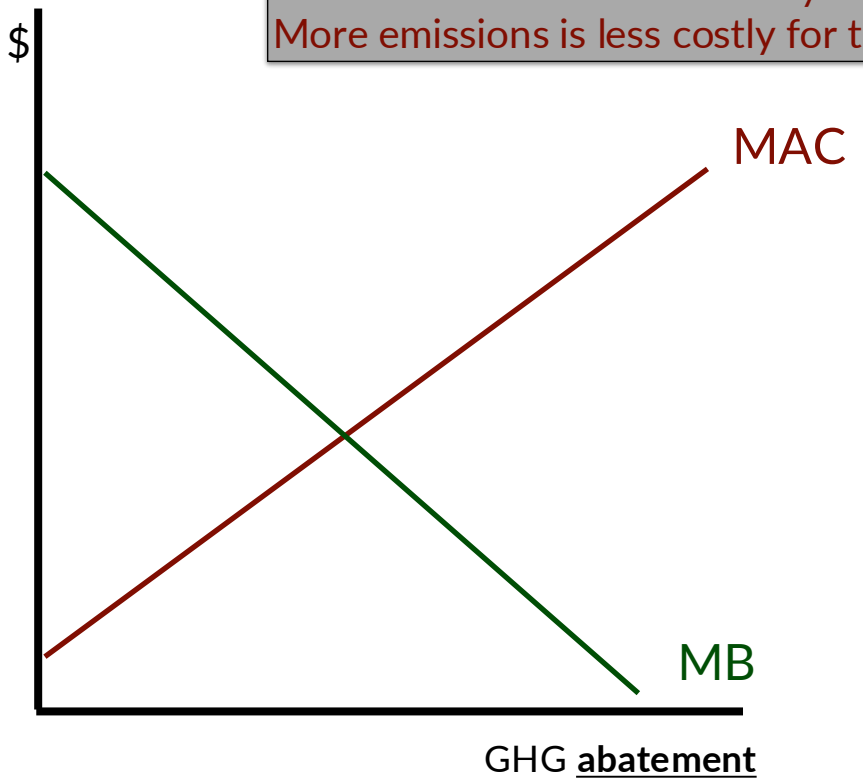
- *Sometimes it's more useful to work with abatement costs. Specifically when we want to think in terms of pollution rather than goods in the output market (e.g., emissions from electricity sector rather than electricity).*

To maximize efficiency (minimize costs), we want all firms to set MAC = MB.

Useful to think in terms of GHG *abatement*

Note: We can plot the horizontal axis in terms of *emissions* or *abatement*.

More abatement is more costly for the firm.
More emissions is less costly for the firm.



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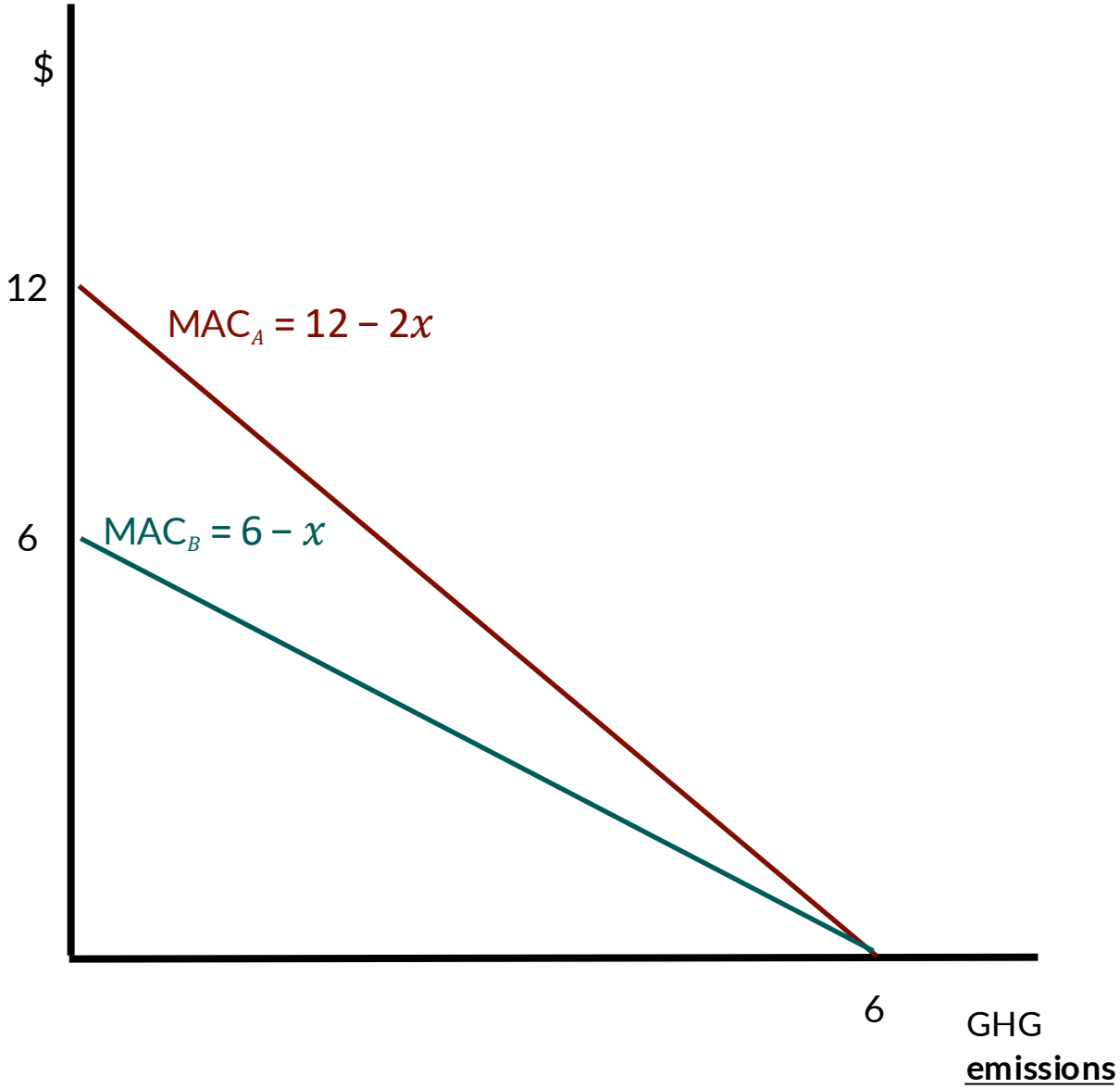
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Performance standards

- Consider two firms (A and B) with different MAC curves.
- 12 units of GHG emissions are generated without regulation.
- Goal: *limit total pollution to 9 units of GHGs.*

- Performance standards set a maximum amount of pollution from each source.
  - E.g., limit each firm to producing 9/n = 4.5 units of GHGs

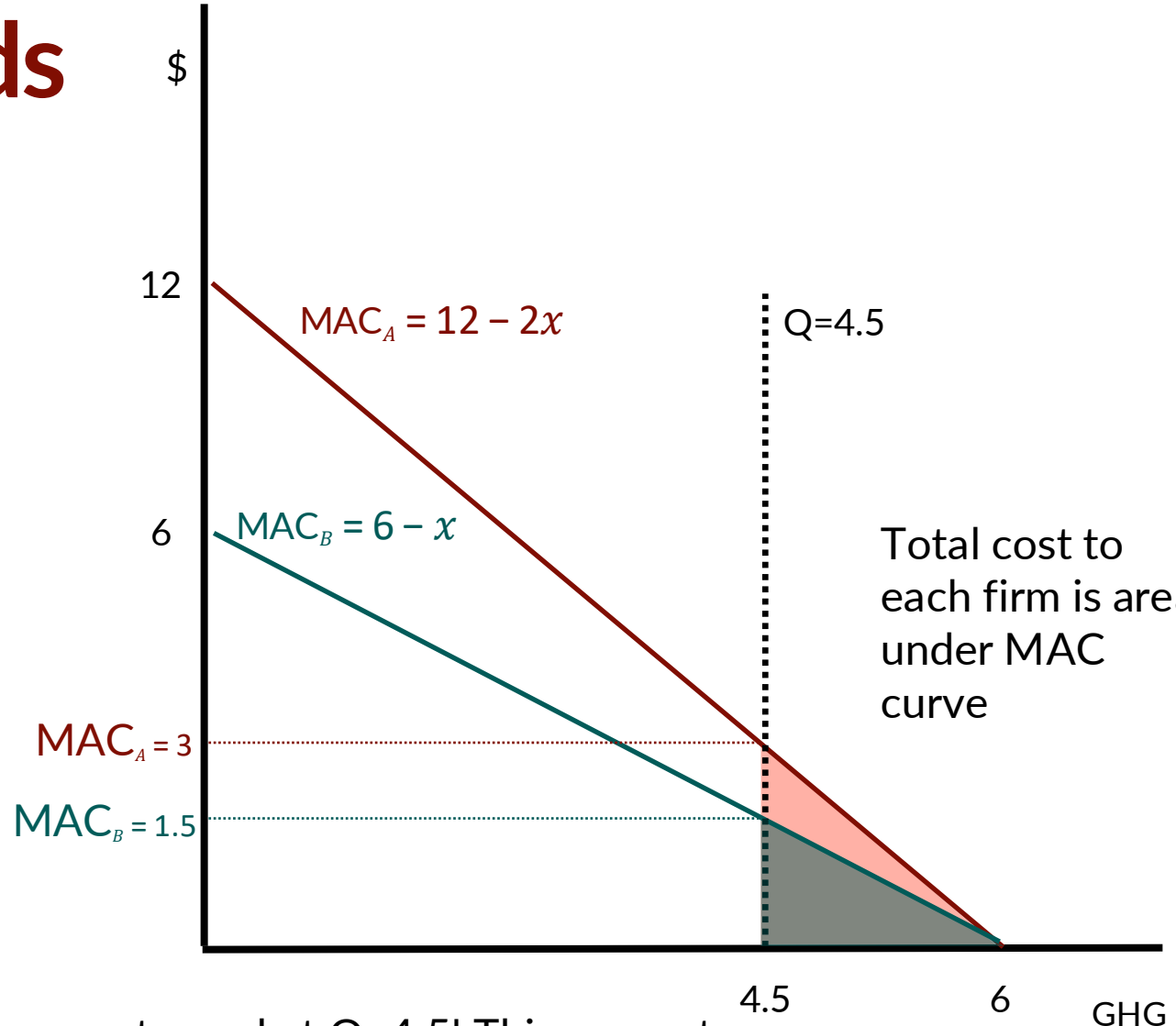


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- Total abatement costs (TAC):
  - $TAC_A = 0.5 \times 1.5 \times 3 = 2.25$
  - $TAC_B = 0.5 \times 1.5 \times 1.5 = 1.125$
  - $TAC_{total} = 3.375$



MACs are not equal at Q=4.5! This suggests that firm B could emit one fewer emission of GHGs more cheaply than firm A.

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Performance standards

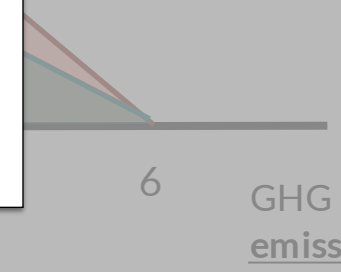
Pros:

- Easy to implement and monitor.

Cons:

- All decision-making rests with the regulator/government
- In general, regulator does not observe MAC curves before setting policy.
  - This means that the regulator does not know who can reduce GHG emissions at lowest cost.
  - Thus, performance standards must be costlier than alternatives.

Total cost to each firm is area under MAC curve

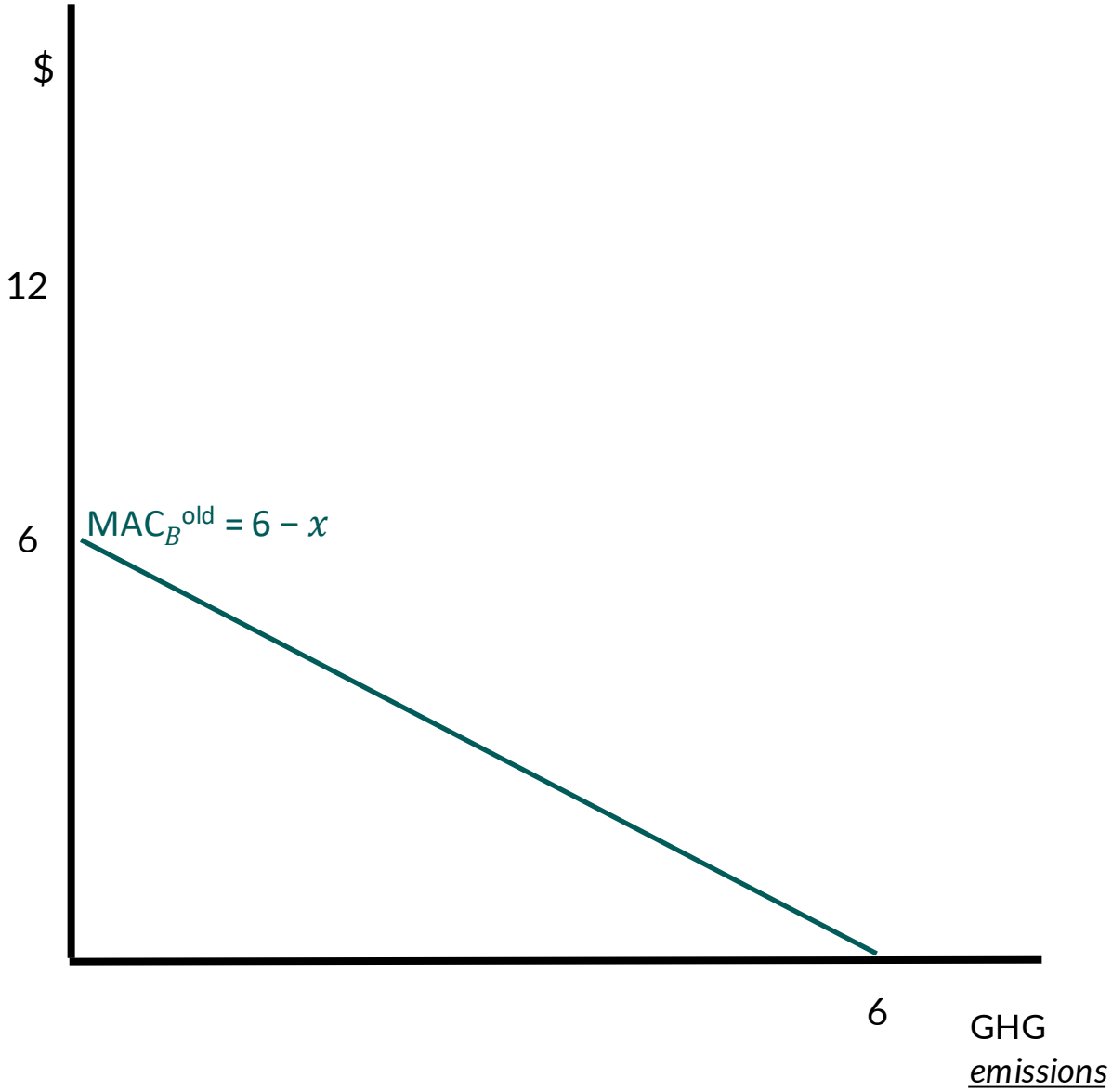


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Technology standards

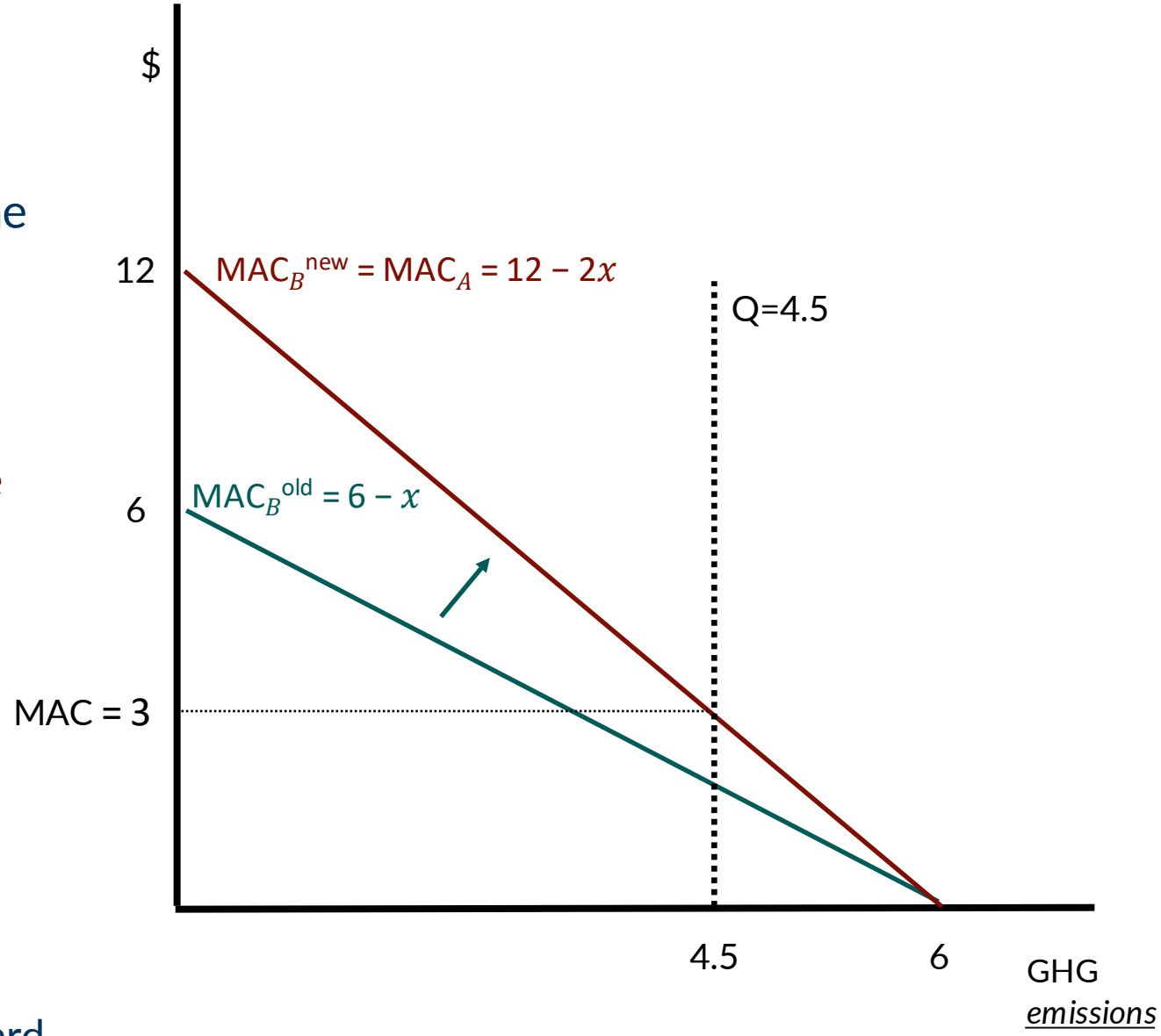
- Technology standards require firm B to use the same technology as firm A
  - The regulator is not choosing quantity of emissions produced
- Changes in technology represent a shift in the MAC curve for firm B.
  - Might not allow firm to use "cheaper" technology.



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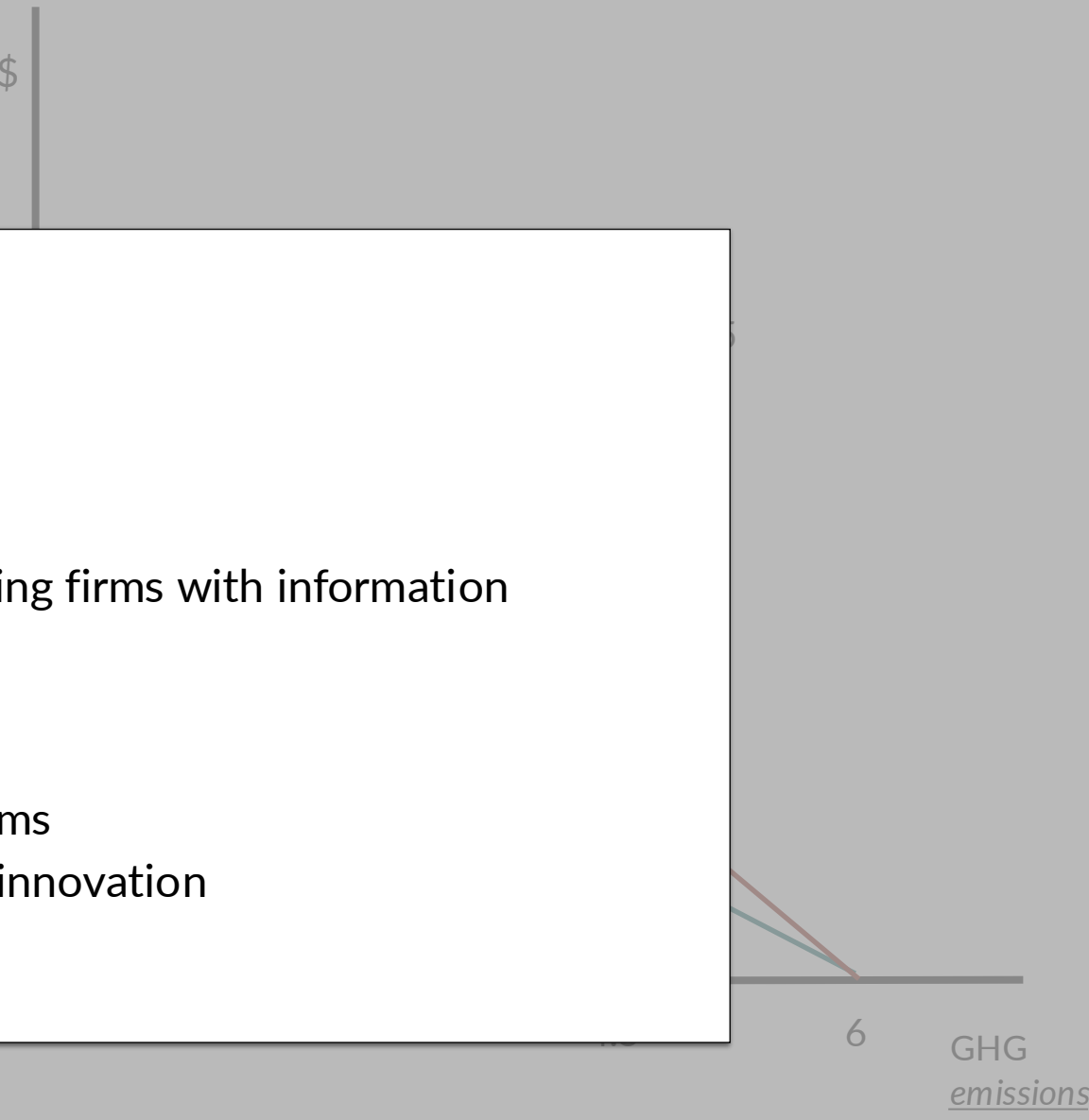
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- Technology standards require firm B to use the same technology as firm A
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- Changes in technology represent a shift in the MAC curve for firm B.
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- To reach Q=4.5, both firms have same MAC curve.
- So, $TAC_A = TAC_B = 0.5 \times 1.5 \times 3 = 2.25$
- $TAC_{total} = 4.5$
  - This was 3.375 under the performance standard
  - Standards on technology increased costs



Technology standards

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- TAC_{total} = 4.5
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Technology standards

Pros:

- May help some under-performing firms with information asymmetry

Cons:

- Can increase costs for some firms
- No incentive for technological innovation

Least-cost allocation

- Same two firms, same goal: 9 units of GHG emissions in total.
- The least-cost allocation equalizes MACs across firms:
  - $12 - 2x_A = 6 - x_B$ and $x_A + x_B = 9$
  - $\Longrightarrow x_A = 5,\ x_B = 4,\ \text{MAC}_A = \text{MAC}_B = 2$
- Firm B has lower abatement costs, so firm B abates more (2 units vs. 1).

Policy	Firm A emits	Firm B emits	TAC_A	TAC_B	$\text{TAC}_{\text{total}}$
Performance standard	4.5	4.5	2.25	1.125	3.375
Technology standard	4.5	4.5	2.25	2.25	4.50
Least cost	**5**	**4**	**1.00**	**2.00**	**3.00**

- Getting firms to internalize a price on emissions of $2/unit produces the "least cost" allocation. Each firm abates until its MAC equals the price on emissions.

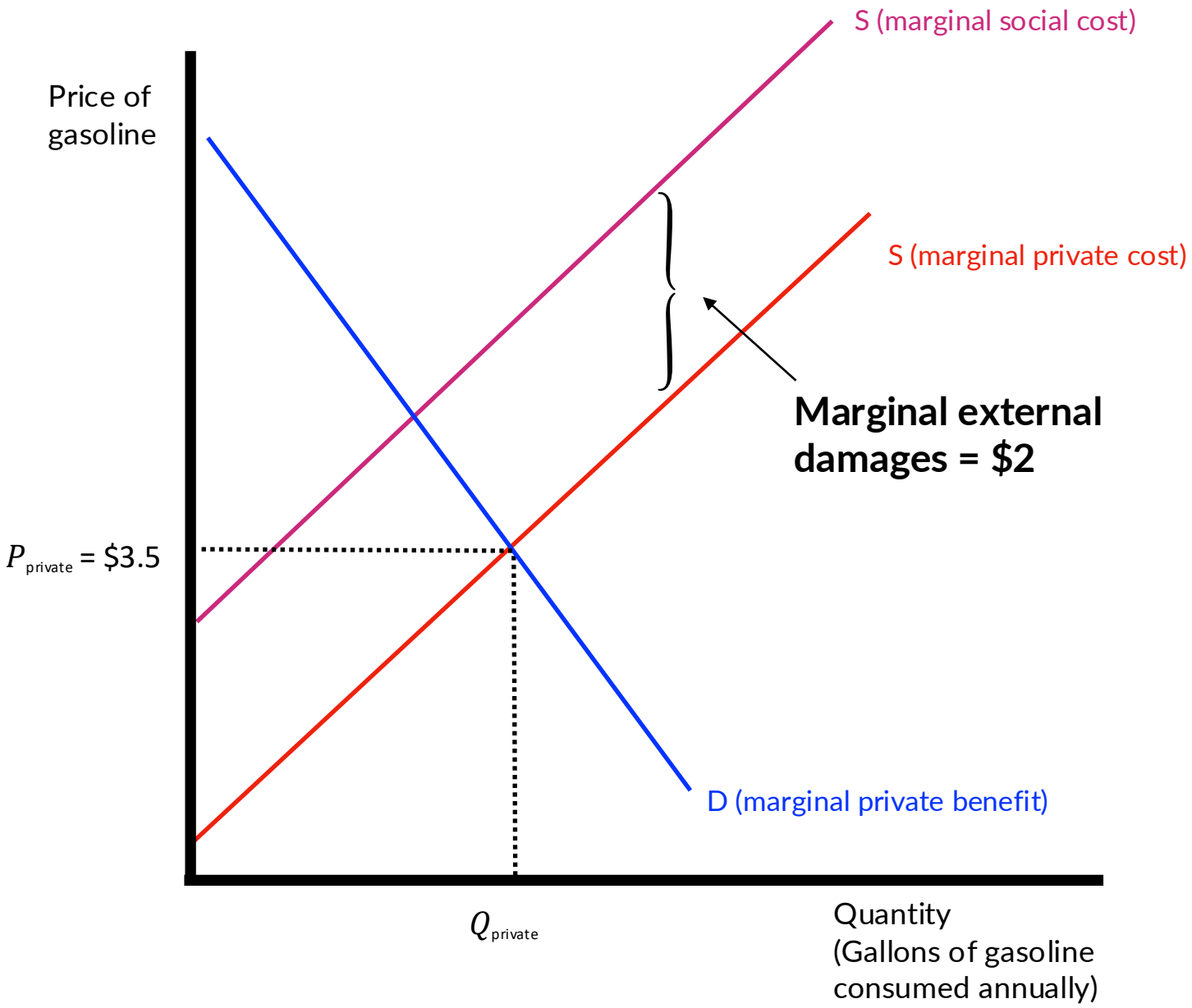
Market-based instruments: Taxes

- Externalities drive a "wedge" between social and private costs.
  - With a negative externality (like GHG emissions), a tax can be used to raise the private marginal cost.
  - With a positive externality (like masks or vaccines), a subsidy can be used to reduce the private cost.

- **Pigouvian (or corrective) taxes:** taxes that are set equal to the marginal damage caused by an externality.
  - Pigouvian taxes *internalize* the externality by putting a price on pollution.

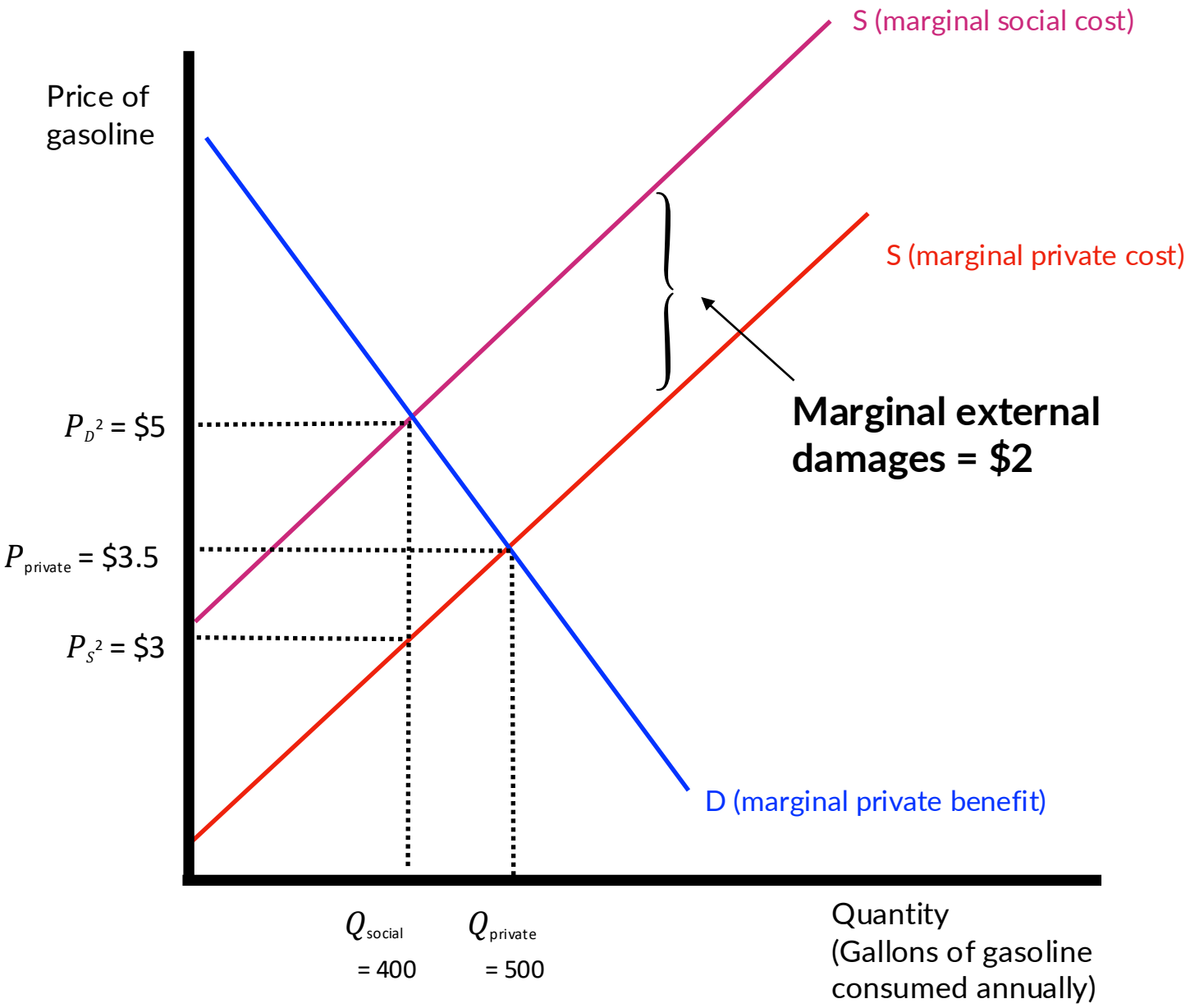
- Corrective taxes act like a "swear jar"
  - https://fordschool.umich.edu/video/2020/catherine-hausman-understanding-carbon-tax

Market-based instruments: Taxes



- A **Pigouvian tax** would *"internalize the external damages"* of gasoline consumption.
  - In this example, marginal external damages are $2 per gallon of gasoline.
- So, setting a tax (τ=$2) would be optimal

Market-based instruments: Taxes

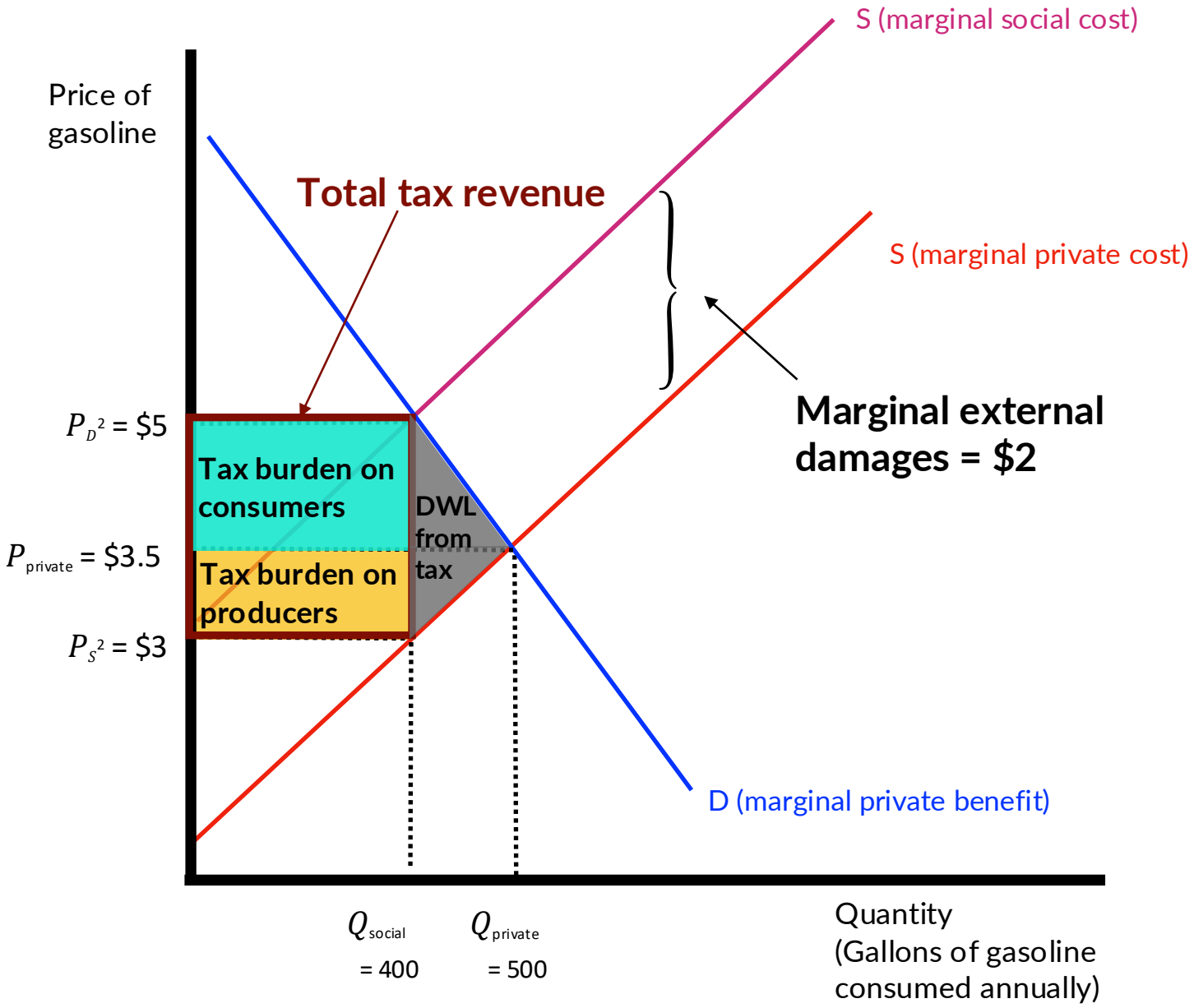


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- Why?
  - Consumers face new price P_{D^2} = $5 and will consume less.
  - Producers sell at new price P_{S^2} = $3 (even though they'd like to sell more at a higher price…)
  - They will arrive at socially optimal consumption/production of gasoline.

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Market-based instruments: Taxes



- $2/gallon Pigouvian tax on gasoline
- **Tax revenue** is generated
  - Total tax revenue = ($5 − $3) × 400 = $800
  - The government can do things with tax revenue!
  - Consumers' **tax burden** is area b/w original price and new price to the left of Q_{social} ($1.50 × 400 = $600).
    - **Tax incidence** is per-unit amount of tax paid:
    - Consumers pay $5–$3.50 = $1.50/gal of the $2/gal tax.
  - Producers' **tax burden** is $0.50 × 400 = $200.
    - Producers pay $0.50/gal of the $2/gal tax.
  - Taxes also generate DWL: DWL from tax = lost consumer and producer surplus (relative to private optimum)

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Market-based instruments: Taxes

Pigouvian taxes

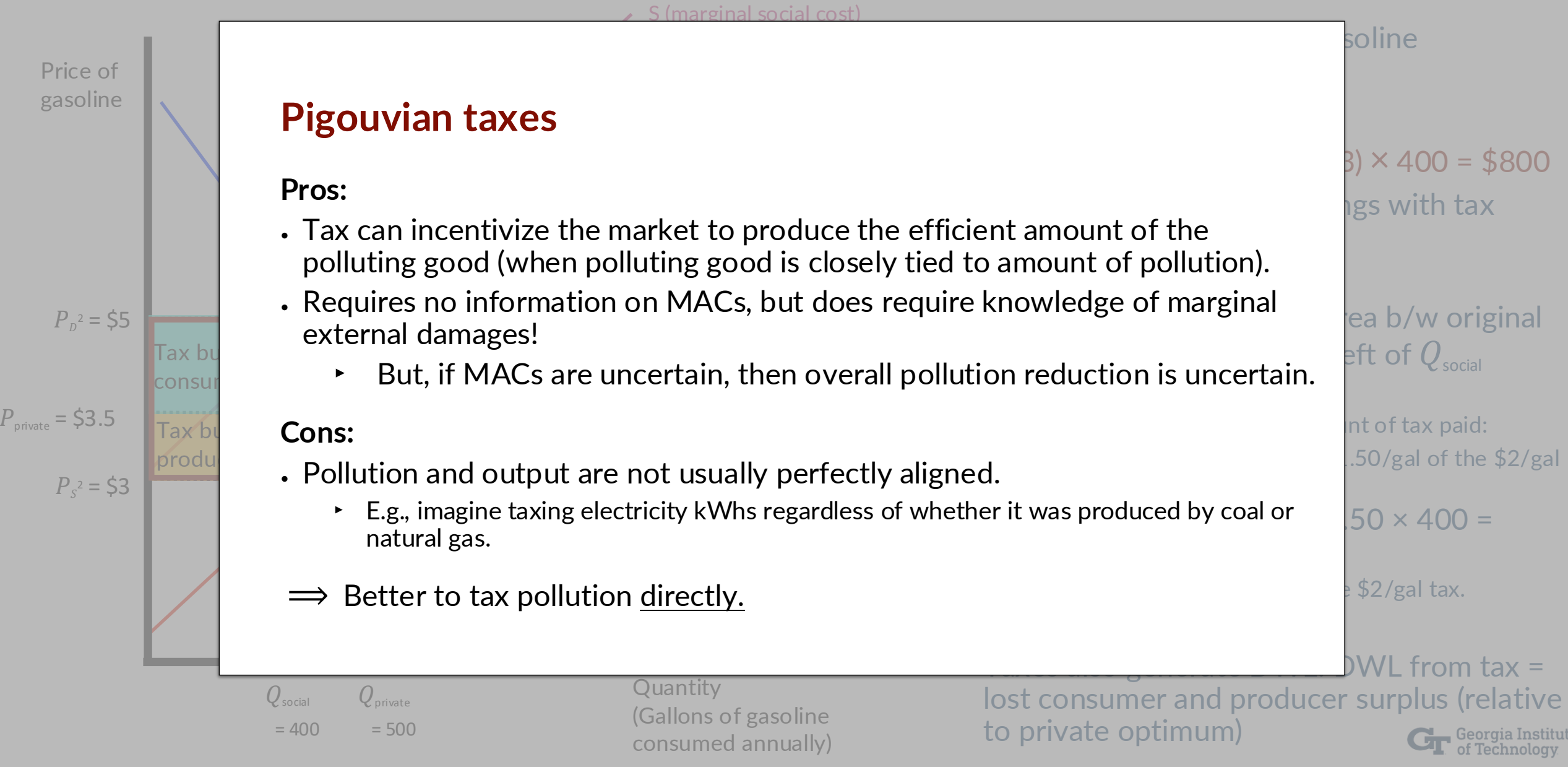
Pros:

- Tax can incentivize the market to produce the efficient amount of the polluting good (when polluting good is closely tied to amount of pollution).
- Requires no information on MACs, but does require knowledge of marginal external damages!
  - But, if MACs are uncertain, then overall pollution reduction is uncertain.

Cons:

- Pollution and output are not usually perfectly aligned.
  - E.g., imagine taxing electricity kWhs regardless of whether it was produced by coal or natural gas.

$\Longrightarrow$ Better to tax pollution directly.

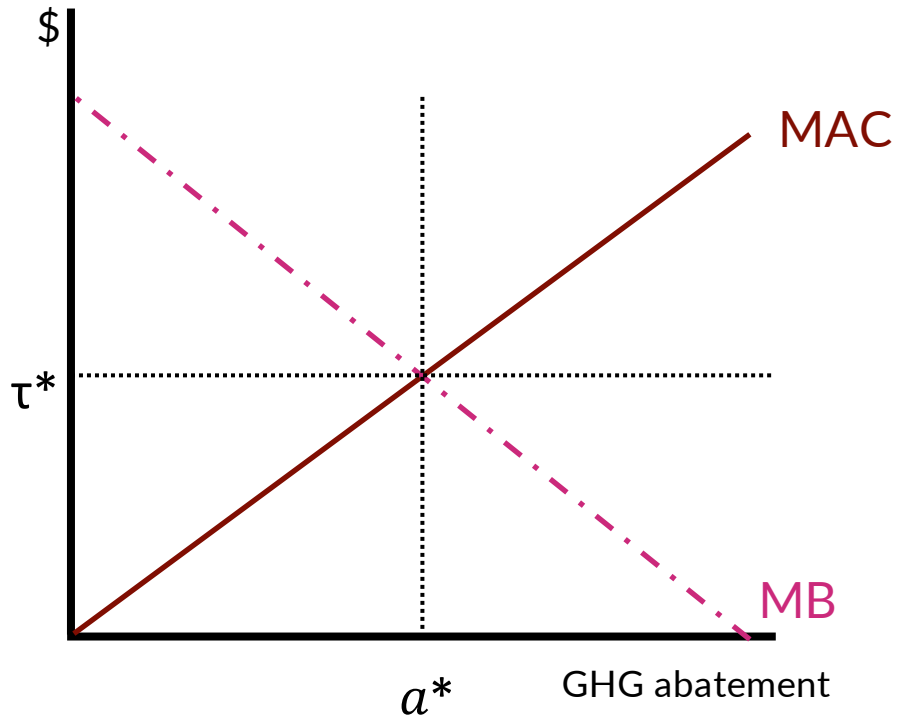


Market-based instruments: Taxes

"Filling in the missing demand curve"

- There is a missing demand curve for abatement
  - GHG abatement is a public good.
  - No one wants to supply a good with a price of zero.
  - The government's role is to "fill in" this demand for abatement.

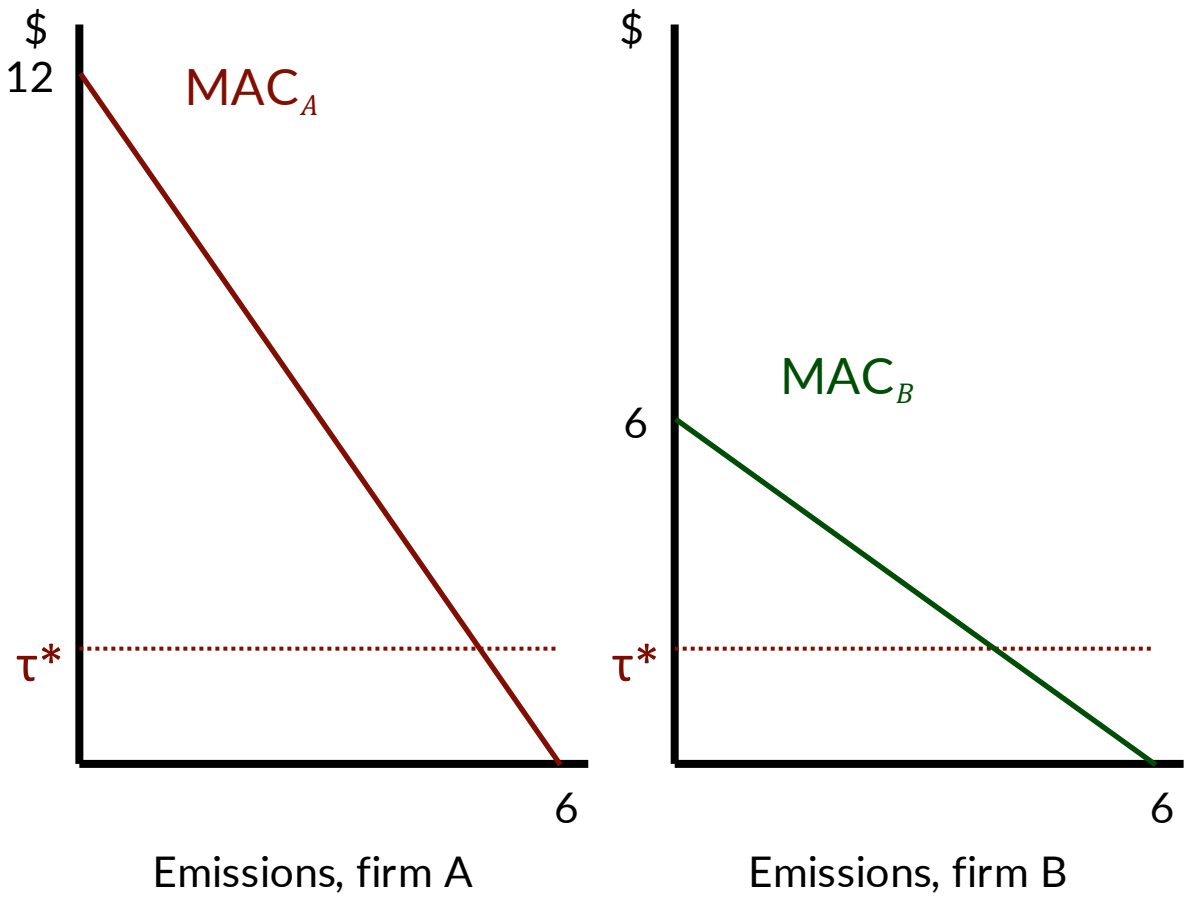
- Gov't can't really set a downward sloping MB curve
  - This would amount to paying each firm different amounts (according to their MACs)
  - Rather, gov't can choose either a uniform tax (τ*) or a fixed quantity (a*) to stand in for the MB curve.
    - Choosing τ* represents a Pigouvian tax
    - Choosing a* represents a cap-and-trade policy (we'll circle back to this!); not a prescriptive standard…
    - Both can achieve the optimal amount of pollution at least cost



Market-based instruments: Taxes

"Filling in the missing demand curve"

- Emissions taxes minimize costs among firms.
- Same two firms as before. Each MAC is plotted against the firm's own emissions:
  - $MAC_A = 12 - 2x_A$ and $MAC_B = 6 - x_B$
- Set the tax (τ^*) equal to marginal external damages. Here, $\tau^* = \$2$.
- Each firm chooses its own emissions.



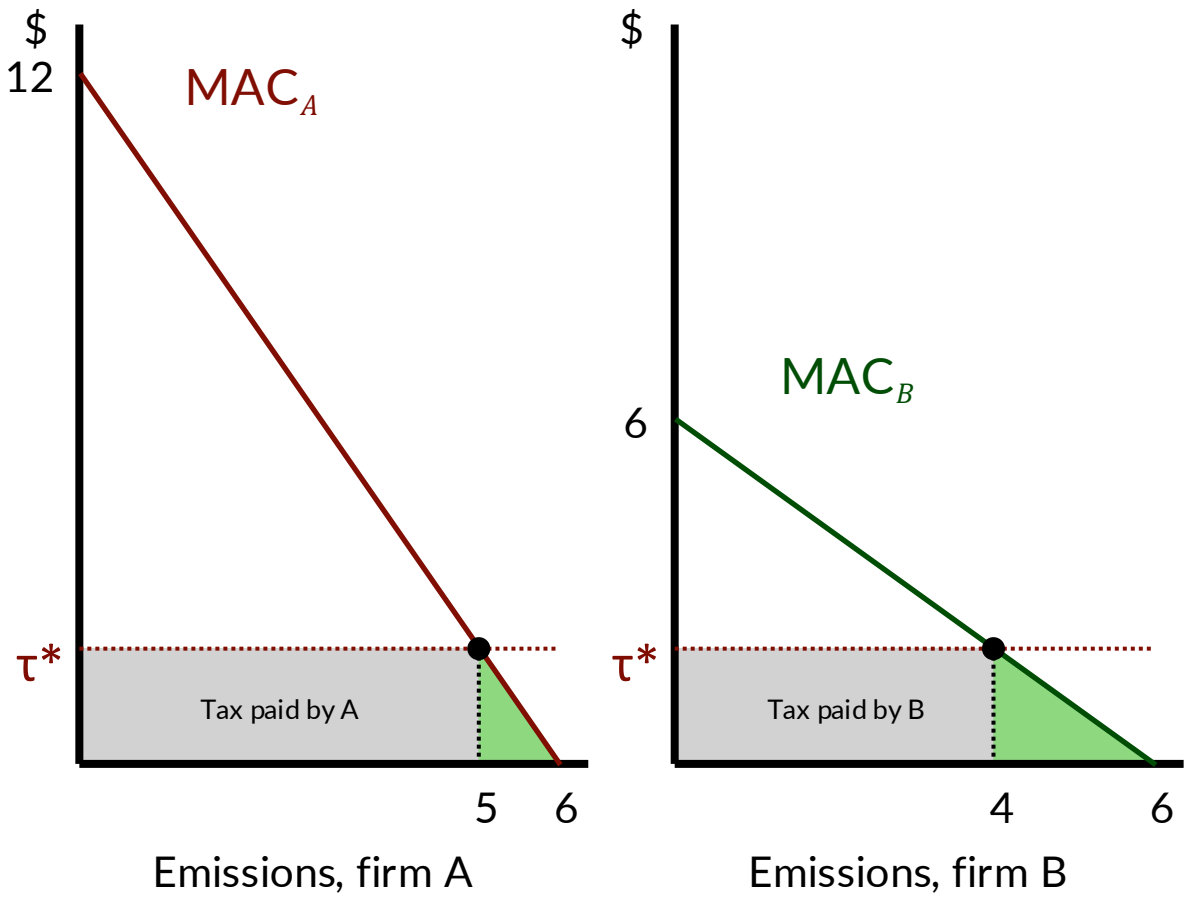
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Market-based instruments: Taxes

"Filling in the missing demand curve"

- Firms set marginal cost of emissions equal to the price of emissions.
  - Without a tax, price = 0.
- Each firm cuts emissions until its MAC equals τ^*:
  - Firm A: $12 - 2x_A = 2 \Longrightarrow x_A = 5$
  - Firm B: $6 - x_B = 2 \Longrightarrow x_B = 4$
- Total emissions are 9 units (3 units of abatement).

- Total abatement cost (green): \$1 + \$2 = \$3.00.
  - This is the least-cost allocation.
- Tax paid on remaining emissions (gray):
  - \$10 + \$8 = \$18.



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Market-based instruments: Taxes

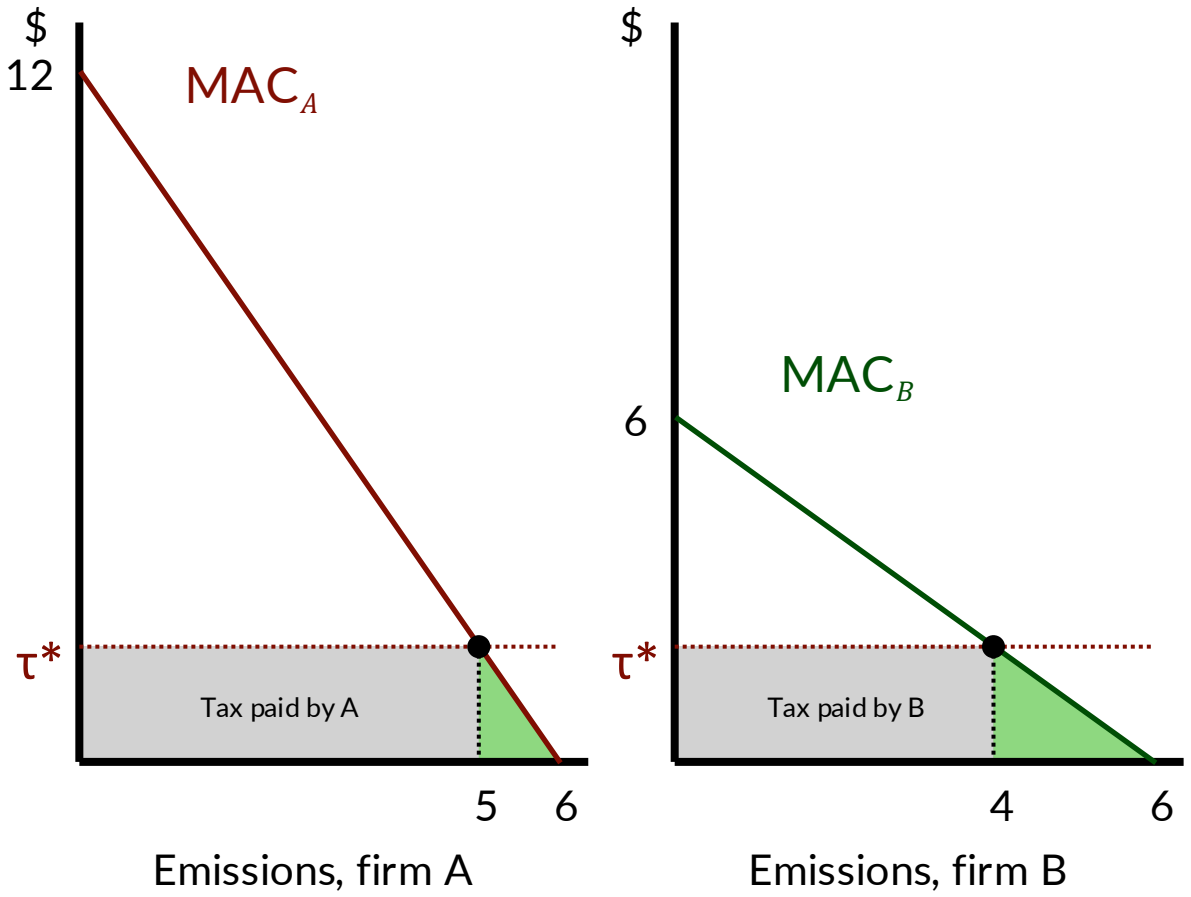
"Filling in the missing demand curve"

Key results:

- Firm B provides more abatement because they have lower abatement costs.
- MACs are equalized at the level of the tax!
- Abatement is achieved at least cost to society.

- Setting $\tau > \tau^*$ results in too much abatement.
- Setting $\tau < \tau^*$ results in too little abatement.
  - Both of these result in welfare loss.

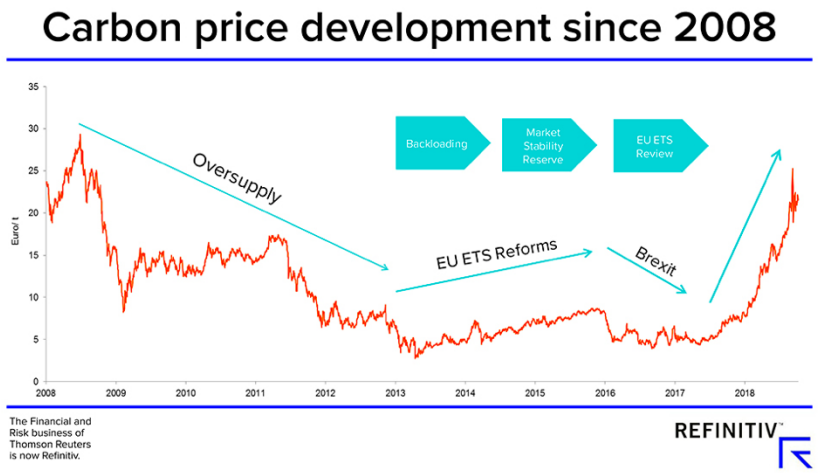
- Taxes might not achieve efficient ***level*** of abatement.
  - Because the regulator does not know each firm's MAC curve before setting a tax, the exact level of abatement/emissions is unclear.
- Taxes will achieve the **least cost reduction** in pollution.

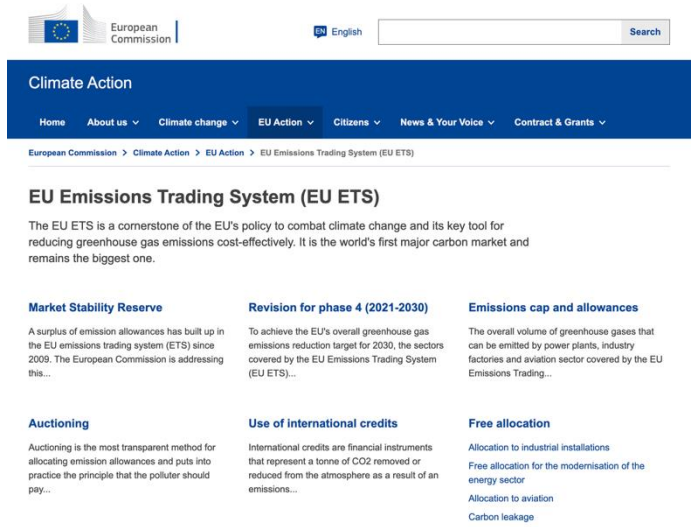


Market-based instruments: Cap-and-trade

- Government provides “permits” to control externalities
  - GHGs can only be emitted if you have a license (i.e., an emissions permit)
    - Gives firms the right to emit
    - Gov’t gives permits away for free or auctions them
  - Only so many permits are issued (limiting total pollution quantity)
  - Trading is allowed between firms to equalize marginal abatement costs across firms.

- Combines aspects of command-and-control and market-based instruments
  - Quantity objective (limit/cap) ← command-and-control
  - Price mechanism (trade) ← economic instrument





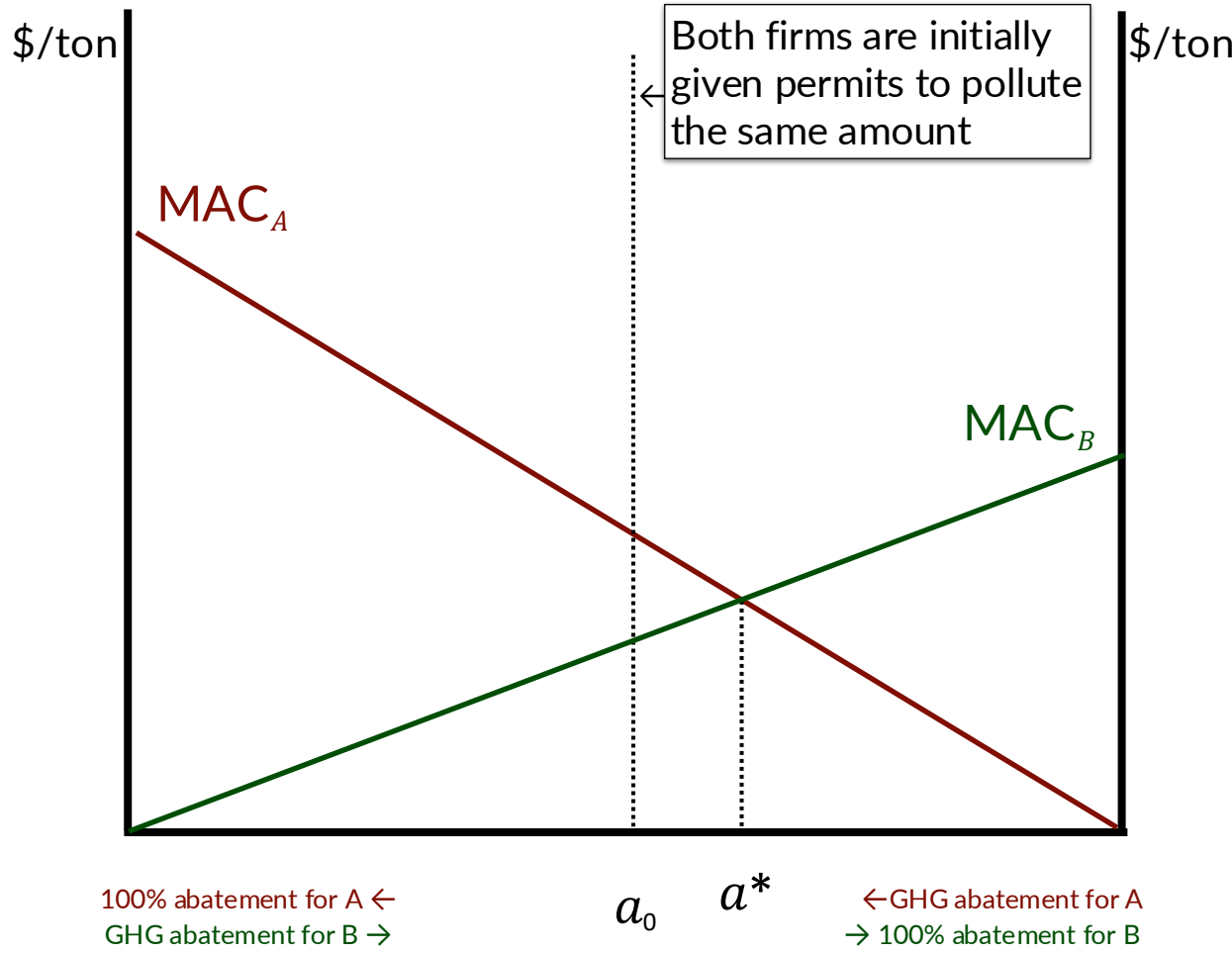
Market-based instruments: Cap-and-trade

"Filling in the missing demand curve"

- Same two firms, now plotted back to back. Firm A's abatement is measured from right to left.
- Imagine the regulator imposes a cap on emissions. The width of the horizontal axis is the total abatement the cap requires.
- Permits are split equally between both firms
- Firms can use the permits (equal to one unit of emissions) to pollute, or they can sell them to other firms.

- Choices for each firm:
  - Reduce emissions to permitted amount
  - Reduce emissions below permitted amount and sell permits.
  - Purchase permits to cover their emissions.

- Firms will buy/sell permits until MACs are equalized.

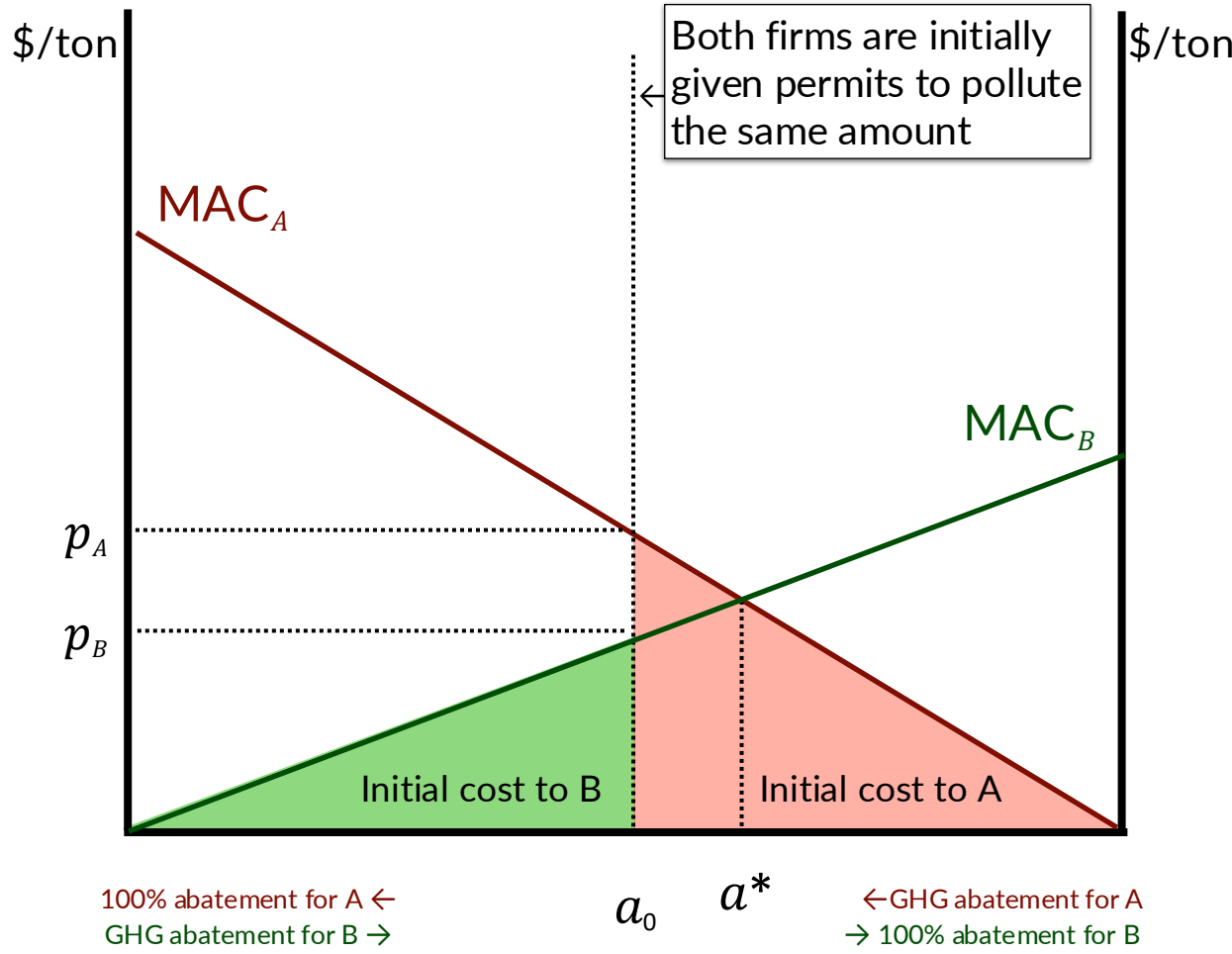


Market-based instruments: Cap-and-trade

"Filling in the missing demand curve"

- Firm B has cheap abatement, so costs at a_0 are relatively small.
- Firm A has expensive abatement, so costs at a_0 are relatively large.

- Firm A should be willing to buy a permit from Firm B because $p_A > p_B$.
  - It costs less for firm A to make this transaction than it does to reduce pollution.



Market-based instruments: Cap-and-trade

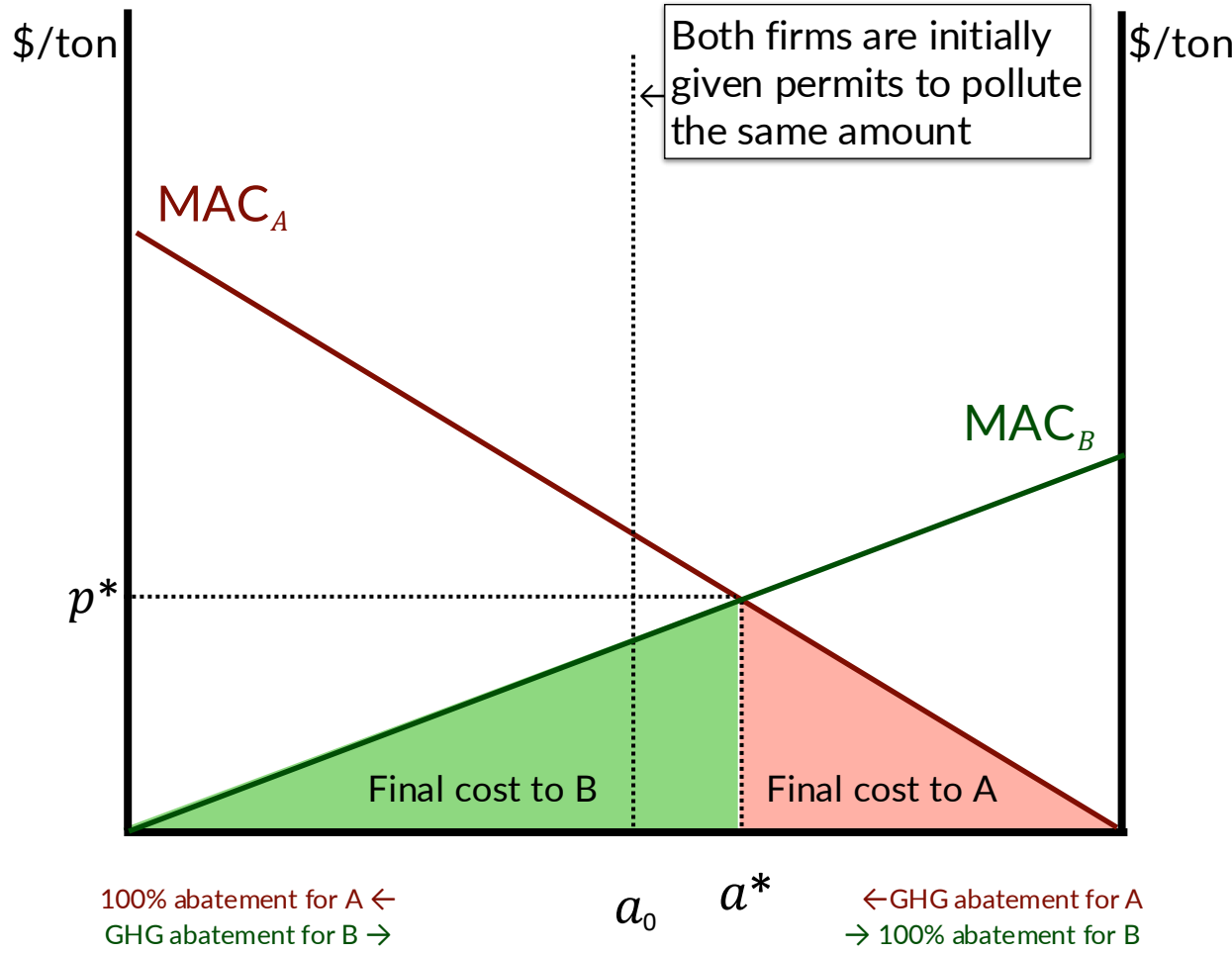
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  - It costs less for firm A to make this transaction than it does to reduce pollution.

- Repeat until MACs are equalized!
  - Results in optimal abatement levels (specifically because that quantity is chosen by regulator)
  - Least-cost reduction in pollution is achieved.

- **Optimal permit price (p^*) equals optimal tax (τ^*)!**



Economists tend to favor market-based solutions…

- "Getting the prices right" sends clear signals to polluting firms.
  - Those with the lowest abatement costs will reduce pollution "first"
- The same price signal rewards firms that find cheaper ways to abate, which encourages technological innovation.
- Carbon taxes generate revenue. Cap-and-trade generates revenue when permits are auctioned (e.g., RGGI, EU ETS).

Next week…

- Monday, 10/26: How would an economist solve climate change? (II)
  - Read: Planet Money, “Sell Me Your Climate Bombs,” and Metcalf (2020).
  - Further reading: Stiglitz (2019). K&O Ch. 9, first half.
- Ghost Emissions round 6 decisions due Sunday, 11:59 PM.

- www.caseyjwichman.com/econ4210#week10